Codd (1970): The Relational Model

Future users of large data banks must be protected from having to know how the data is organized in the machine — the internal representation.

A prompting service which supplies such information is not a satisfactory solution. Activities of users at terminals and most application programs should remain unaffected when the internal representation of data is changed.

Changes in data representation will often be needed as a result of changes in query, update, and report traffic and natural growth in the types of stored information.

Existing noninferential, formatted data systems provide users with tree-structured files or slightly more general network models of the data.

In Section 1, inadequacies of these models are discussed. A model based on n-ary relations, a normal form for database relations, and the concept of a universal data sublanguage are introduced.

Stonebraker & Rowe (1986): The Design of POSTGRES

This paper presents the preliminary design of a new database management system, called POSTGRES, that is the successor to the INGRES relational database system.

The main design goals of the new system are to provide better support for complex objects, to provide user extensibility for data types, operators and access methods, and to provide facilities for active databases (i.e., alerters and triggers) and inferencing including forward- and backward-chaining.

POSTGRES makes as few changes as possible to the relational model. As a result, the query language QUEL is only slightly altered to become POSTQUEL.

Many of the most important changes in POSTGRES occur below the query language interface, in the implementation of data types, access methods, and the transaction system.

Furthermore, POSTGRES is designed to run on a shared memory multiprocessor and include a novel storage system that uses an append-only design with garbage collection of obsolete tuples.

PostgreSQL: From Berkeley to Global Open Source

The object-relational database management system now known as PostgreSQL is derived from the POSTGRES package written at the University of California at Berkeley.

With over two decades of development behind it, PostgreSQL is now the most advanced open-source database available anywhere, supporting a large part of the SQL standard and offering many modern features.

Some of these features are complex queries, foreign keys, triggers, updatable views, transactional integrity, and multiversion concurrency control.

Also, PostgreSQL can be extended by the user in many ways, for example by adding new data types, functions, operators, aggregate functions, index methods and procedural languages.

And because of the liberal license, PostgreSQL can be used, modified, and distributed by anyone free of charge for any purpose, be it private, commercial, or academic.

Modern Era: Vector Search and AI Workloads

The pgvector extension adds vector data types and similarity-search operators to PostgreSQL, enabling semantic search and retrieval-augmented generation directly inside the database.

Vector embeddings are produced by language models such as the OpenAI text-embedding-3-small model, which returns 1536-dimensional float vectors for arbitrary input text.

Approximate-nearest-neighbor indexes (HNSW and IVFFlat) allow sub-linear lookup over millions of vectors with controllable recall.

Combining traditional relational queries with vector similarity makes PostgreSQL a unified store for both structured business data and unstructured AI features.

This is the foundation that pg_aidb extends with managed pipelines, model registry, and the dual-mode synchronous/asynchronous API surface.